Laboratory 02 Report: Dataset Analysis

Machine Learning and Pattern Recognition

1 Theory Section: Iris Dataset

1.1 Loading the Dataset

The Iris dataset was loaded into a data matrix $D \in \mathbb{R}^{4\times150}$, where each column is one sample and each row is one feature. The label vector L has 150 entries and maps Iris-setosa to 0, Iris-versicolor to 1, and Iris-virginica to 2. This organization is useful because feature-wise statistics can be computed along rows, while class masks can select sample columns.

1.2 Visualizing the Features

Normalized histograms were generated for all four Iris features and scatter plots were generated for every pair of features. The first two features, sepal length and sepal width, show substantial overlap between classes. Petal length and petal width are much more informative: Iris-setosa is clearly separated from the other classes, while Iris-versicolor and Iris-virginica still overlap partially.

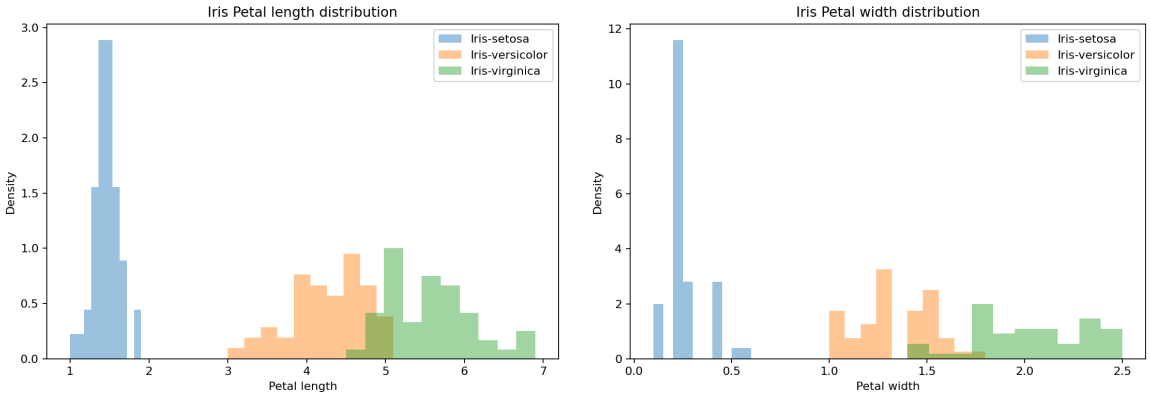


Figure 1: Iris petal-feature histograms. Setosa is well separated from the other classes.

1.3 Dataset Mean, Covariance, Variance, and Standard Deviation

The empirical mean was computed as $\mu = \frac{1}{N}\sum_i x_i$ and the covariance matrix as $C = \frac{1}{N} D_C D_C^T$, where D_C is the centered data matrix. The global feature statistics are shown in Table 1.

Feature	Mean	Variance	Std. dev.
Sepal length	5.843	0.681	0.825
Sepal width	3.057	0.189	0.434
Petal length	3.758	3.096	1.759
Petal width	1.199	0.577	0.760

Table 1: Global Iris feature statistics.

1.4 Per-Class Comparison

The per-class means confirm that petal features provide the strongest class separation. Setosa has mean petal length 1.462 and mean petal width 0.246, while Versicolor has 4.260 and 1.326 and Virginica has 5.552 and 2.026. Sepal features are less discriminative because their class means are closer and their distributions overlap more. Variances are also smaller for Setosa petal features, reinforcing the compact cluster visible in the plots.

2 Project Section: Fingerprint Spoofing Dataset

The project dataset used in this report is `LAB02/data/project/trainData.txt`. It contains 6000 samples, 6 features, and binary labels: 0 for fake fingerprints and 1 for genuine fingerprints.

Analysis of features 1 and 2

What do you observe?

The two classes ("Fake" and "Genuine") occupy distinct regions but are not perfectly separable. Observing the scatter plot (`features_1_2_scatter.png`), the "Genuine" class (label 1) generally clusters towards the top-left (lower Feature 1, higher Feature 2), while the "Fake" class (label 0) clusters towards the bottom-right (higher Feature 1, lower Feature 2).

Do the classes overlap? If so, where?

Yes, the classes overlap significantly. According to the computed data in `lab02_project_feature_overlap_rang`

- **Feature 1** overlaps between -1.16104 and 2.14620 (an overlap width of 3.30724).
- **Feature 2** overlaps between -0.61203 and 1.25841 (an overlap width of 1.87044).

Do the classes show similar mean for the first two features?

No, the means are distinctly different, reflecting the class separation seen in the scatter plots. Extracted from `lab02_project_statistics.csv`:

- **Feature 1:** The "Fake" mean is 0.609, while the "Genuine" mean is much lower at -0.655.
- **Feature 2:** The "Fake" mean is -0.198, while the "Genuine" mean is higher at 0.704.

Are the variances similar for the two classes?

The variances are in a similar general range, though not identical. According to `lab02_project_statistics.cs`

- **Feature 1:** The variance for "Fake" (0.370) is slightly higher than for "Genuine" (0.316).
- **Feature 2:** The variance for "Fake" (0.264) is noticeably lower than for "Genuine" (0.395), which corresponds to the histograms where the "Fake" distribution appears slightly narrower.

How many modes are evident from the histograms?

Based on `feature_1_histogram.png` and `feature_2_histogram.png`, exactly one mode is evident for each class. Both the "Fake" and "Genuine" distributions exhibit a single, distinct peak (a unimodal distribution) for both Feature 1 and Feature 2.

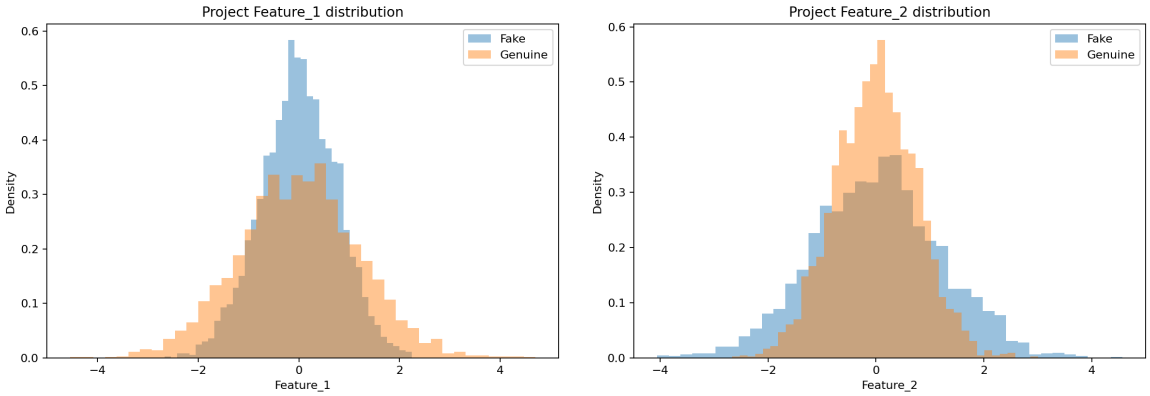


Figure 2: Project features 1 and 2. The classes have similar means but different variances.

Analysis of features 3 and 4

What do you observe?

Observing the scatter plot (`features_3_4_scatter.png`), we can analyze the distribution of the two classes ("Fake" and "Genuine"). Given that the mean values for the two classes have opposite signs for both features, the classes appear to occupy distinct regions (specifically, opposite quadrants), although they are somewhat mixed in the center. The "Genuine" class tends to have higher values for Feature 3 and lower values for Feature 4, while the "Fake" class shows the exact opposite behavior.

Do the classes overlap? If so, where?

Yes, the classes exhibit overlap. According to the computed data in `lab02_project_feature_overlap_ranges.c`

- **Feature 3** overlaps between -2.0146 and 1.9045 (an overlap width of 3.9190).
- **Feature 4** overlaps between -1.6829 and 1.8171 (an overlap width of 3.5000).

Do the classes show similar mean for these two features?

No, the means are distinctly different and exhibit roughly opposite signs. Extracted from `lab02_project_statistics.csv`:

- **Feature 3:** The "Fake" mean is -0.6809, while the "Genuine" mean is 0.6652.
- **Feature 4:** The "Fake" mean is 0.6708, while the "Genuine" mean is -0.6642.

Are the variances similar for the two classes?

Yes, the variances are relatively similar and almost identical. According to `lab02_project_statistics.csv`:

- **Feature 3:** The variance for "Fake" is 0.5500, and for "Genuine" it is 0.5489.
- **Feature 4:** The variance for "Fake" is 0.5360, and for "Genuine" it is 0.5533.

How many modes are evident from the histograms?

Based on `feature_3_histogram.png` and `feature_4_histogram.png`, we can observe the following modes:

- **Feature 3:** The "Fake" distribution exhibits one mode, and the "Genuine" distribution exhibits one mode.
- **Feature 4:** The "Fake" distribution exhibits one mode, and the "Genuine" distribution exhibits one mode.

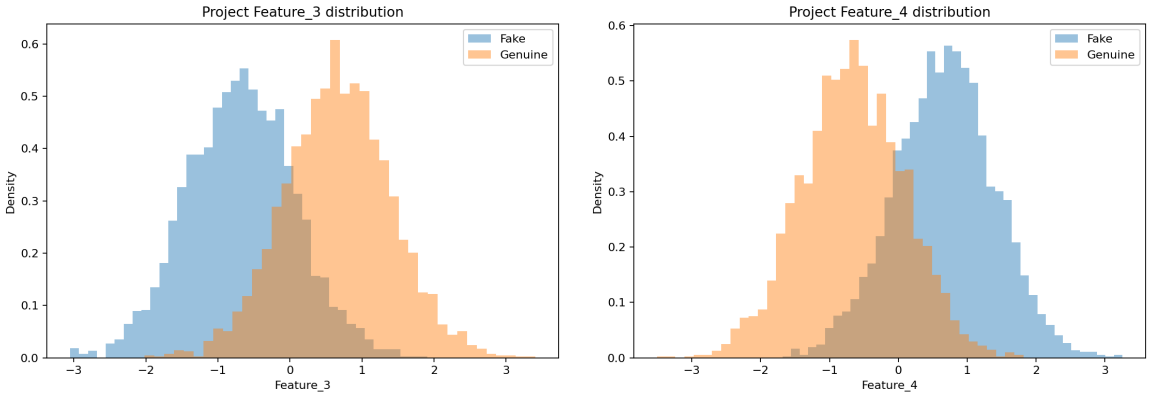


Figure 3: Project features 3 and 4. The classes mainly differ by mean.

Analysis of features 5 and 6

What do you observe?

Observing the scatter plot (`features_5_6_scatter.png`), the classes exhibit a highly structured, non-linear distribution. The "Fake" class forms a cross-like shape with clusters situated along the axes, while the "Genuine" class forms a ring of clusters situated in the four corners (quadrants) of the plot. Because of this geometry, the classes are heavily mixed in a way that a simple linear decision boundary would fail to separate them effectively.

Do the classes overlap? If so, where?

Yes, the classes overlap extensively across almost the entire range of values. According to the computed data in `lab02_project_feature_overlap_ranges.csv`:

- **Feature 5** overlaps between -2.0731 and 1.9972 (an overlap width of 4.0703).
- **Feature 6** overlaps between -2.0130 and 2.1906 (an overlap width of 4.2035).

Do the classes show similar mean for these two features?

Yes, the means are very similar and both are centered extremely close to zero. Extracted from `lab02_project_statistics.csv`:

- **Feature 5:** The "Fake" mean is 0.0280, while the "Genuine" mean is -0.0417.
- **Feature 6:** The "Fake" mean is -0.0058, while the "Genuine" mean is 0.0239.

Are the variances similar for the two classes?

No, the variances show a significant difference, with the "Genuine" class having nearly double the variance of the "Fake" class. According to `lab02_project_statistics.csv`:

- **Feature 5:** The variance for "Fake" is 0.6801, and for "Genuine" it is 1.3178.
- **Feature 6:** The variance for "Fake" is 0.7050, and for "Genuine" it is 1.2870.

How many modes are evident from the histograms?

Based on `feature_5_histogram.png` and `feature_6_histogram.png`, we can observe the following modes:

- **Feature 5:** The "Fake" distribution exhibits one mode (centered at 0), and the "Genuine" distribution exhibits two distinct modes.
- **Feature 6:** The "Fake" distribution exhibits one mode (centered at 0), and the "Genuine" distribution exhibits two distinct modes.

How many clusters can you notice from the scatter plots for each class?

Based on the visual distribution in `features_5_6_scatter.png`, we can notice:

- **Fake class:** 4 distinct clusters (located radially along the x and y axes).
- **Genuine class:** 4 distinct clusters (located in the four diagonal quadrants).

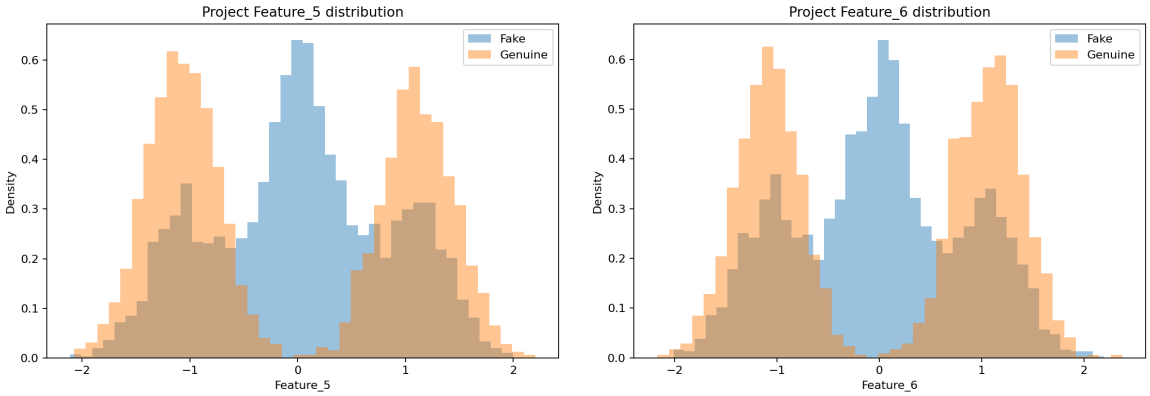


Figure 4: Project features 5 and 6. The classes overlap strongly, with larger genuine variance.